

Solutions Set Theory II

1. Let $S = \{\text{Homer, Marge, Bart, Lisa, Maggie}\}$. Enumerate the following relations.

(a) "is a sibling of"

$$R = \{(B, L), (B, Ma), (L, B), (L, Ma), (Ma, B), (Ma, L)\}$$

(M) for Marge and (Ma) for Maggie.

(b) "is married to"

$$R = \{(H, M), (M, H)\}$$

(c) "is taller than"

$$R = \{(M, H), (M, B), (M, L), (M, Ma), (H, B), (H, L), (H, Ma), \\ (B, L), (B, Ma), (L, Ma)\}$$

(d) "is older than"

$$R = \{(H, M), (H, B), (H, L), (H, Ma), (M, B), (M, L), (M, Ma), \\ (B, L), (B, Ma), (L, Ma)\}$$

2. Let $S = \{1, 2, 3, 4\}$. Graph the following relations.

(a) =

4				.
3			.	
2		.		
1	.			
=	1	2	3	4

(b) <

4				
3				.
2			.	.
1		.	.	.
<	1	2	3	4

- (c) $R = \{(1, 1), (2, 1), (2, 2), (3, 3), (4, 3), (4, 4)\}$

4			.	.
3			.	
2	.	.		
1	.			
R	1	2	3	4

3. Determine the following sets.

- (a) The upper contour set of Lisa in 1c.

$$S = \{(B, L), (M, L), (H, L)\}$$

- (b) The lower contour set of Bart in 1d.

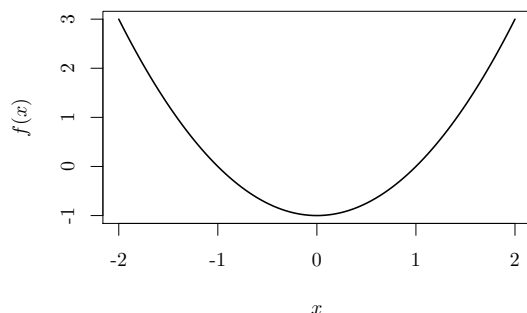
$$S = \{(M, B), (H, B)\}$$

- (c) The upper contour set of 2 in 2b.

$$S = \{(1, 2)\}$$

- (d) Let $S = \mathbb{R}$. Determine the upper contour set of $x \in S$ given the relation $R = \{(x, y) \in S \times S \mid x^2 - 1 = y\}$. Graph the binary relation. Is the upper contour set convex or concave or none of both?

$$f(x) = x^2 - 1$$



The upper contour set is the area above the graph. It is convex.

4. Check whether the following relations are reflexive, irreflexive, transitive, complete, symmetric, antisymmetric and/or asymmetric. Also check whether they are a weak, strict, weak partial or strict partial order (or none of those).

(a) $\leq$

(b) $<$

(c) $=$

(d) $R_1 = \{(1, 1), (1, 2), (1, 3), (1, 4)\}$ ₂

(e) $R_2 = \text{"was born before"}$

(f) $R_3 = \{(a, a), (a, b)\}$

•	ref	irr	trans	com	sym	anti	asym	weak	strict	partial
$\leq$	✓	•	✓	✓	•	✓	•	✓		✓
$<$	•	✓	✓	✓	•	✓	✓		✓	✓
$=$	✓	•	✓	✓	✓	✓	•	✓		✓
R_1	✓	•	✓	•	•	✓	✓			✓
R_2	•	✓	✓	•	•	✓	•			✓
R_3	✓	•	✓	•	•	✓	•			✓

5. Let $S = \{1, 2, 3\}$. Show by example that...

(a) ...if R is asymmetric, it is also antisymmetric.

An asymmetric relation, say $>$, is:

$$R_1 = (2, 1), (3, 2), (3, 1)$$

For antisymmetry it must hold that $\forall x, y \in S, [(xRy) \wedge (yRx)] \implies (x = y)$. Since the hypothesis is always false, the statement is always true and thus, the relation is antisymmetric.

(b) ...if R is asymmetric, it is also irreflexive.

Consider the same relation R_1 . Irreflexivity requires that $\nexists x \in S, xRx$ which is the case.

(c) ...if R is irreflexive and transitive, it is also asymmetric. Consider the irreflexive relation

$$R_2 = \{(1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2)\}.$$

To meet also transitivity, the relation reduces to

$$R_3 = \{(1, 2), (1, 3), (2, 3)\}.$$

It can be seen that this relation is also asymmetric.

(d) ...if R is symmetric and antisymmetric, it is also transitive. As an example consider the relation $=$ which is symmetric and also antisymmetric.

$$R_4 = \{(1, 1), (2, 2), (3, 3)\}.$$

This relation is also transitive since the hypothesis is always false and therefore the statement is true.

As another example consider the relation $\neq$. This relation is symmetric, but it is not antisymmetric and therefore not necessarily transitive.

6. Let $A = \{\{a\}, \{b\}, \{a, b\}\}$. Let $B = \mathcal{P}(A) \setminus \emptyset$, where $\mathcal{P}(A)$ is the *power set*, the set of all subsets of A . Define a binary relation $R \equiv \subseteq$.

- (a) Explicitly enumerate B , and state its cardinality.

The power set of A minus the empty set is

$$B = \{\{\{a\}\}, \{\{b\}\}, \{\{a, b\}\}, \{\{a\}, \{b\}\}, \{\{a\}, \{a, b\}\}, \{\{b\}, \{a, b\}\}, \{\{a\}, \{b\}, \{a, b\}\}\}$$

Its cardinality is 7 ($2^3 - 1$).

- (b) Prove whether R is a weak order on B .

A weak order requires reflexivity, completeness and transitivity. Consider the completeness requirement. $\{\{a\}\} \subseteq \{\{b\}\}$ is false. The reverse, $\{\{b\}\} \subseteq \{\{a\}\}$ is also false. Thus, the relation is not complete and therefore R is not a weak ordering on B .

- (c) Prove whether R is a partial order on B .

A partial ordering requires reflexivity, transitivity and antisymmetry. Reflexivity requires that $\forall x \text{ in } S, xRx$. This is clearly the case, since $\subseteq$ includes equality and thus $\forall x \text{ in } B, x = x$. Transitivity is also given: If $x \subseteq y$ and $y \subseteq z$, then $x \subseteq z$. For antisymmetry, it needs to hold that (xRy) and (yRx) imply $x = y$. The only case for which the premise can hold for the relation $\subseteq$ is if $x = y$. Thus, $\subseteq$ is a partial order on B .

- (a) Let R_1 and R_2 be transitive relations on a set S . Does it follow that $R_1 \cup R_2$ is transitive?

No. A counterexample: $S = \{1, 2\}$. $R_1 = '>' = \{(2, 1)\}$ and $R_2 = '<' = \{(1, 2)\}$. And $R_1 \cup R_2 = \{(1, 2), (2, 1)\}$ is not transitive.

- (b) Let R_1 and R_2 be transitive relations on a set S . Does it follow that $R_1 \cap R_2$ is transitive?

Yes. Assume R_1 and R_2 are both transitive and let $(a, b), (b, c) \in R_1 \cap R_2$. Then $(a, b), (b, c) \in R_1$ and $(a, b), (b, c) \in R_2$. It is given that both R_1 and R_2 are transitive, so $(a, c) \in R_1$ and $(a, c) \in R_2$. Therefore $(a, c) \in R_1 \cap R_2$. This shows that for arbitrary $(a, b), (b, c) \in R_1 \cap R_2$ we have $(a, c) \in R_1 \cap R_2$. Thus $R_1 \cap R_2$ is transitive.